1

(a)

Solution. We are given

$$a_n = \frac{n^3 - 7}{n^4 + 3}.$$

Idea: Compare dominant powers. The numerator is order n^3 and the denominator is order n^4, so we expect $a_n \to 0$.

Step 1: Divide by the highest power n^4

$$a_n = \frac{n^3 - 7}{n^4 + 3} = \frac{n^3}{n^4} \cdot \frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}} = \frac{1}{n} \cdot \frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}}.$$

Step 2: Take limits of each factor - $\frac{1}{n} \to 0$, - $1 - \frac{7}{n^3} \to 1$, - $1 + \frac{3}{n^4} \to 1$, so by limit laws,

$$\frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}} \longrightarrow 1 \quad \Longrightarrow \quad a_n \;=\; \frac{1}{n} \cdot \left(\frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}} \right) \;\longrightarrow\; 0 \cdot 1 \;=\; 0.$$

Optional ε-proof (quick bound) For $n \geq 1$,

$$\left| \frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}} \right| \leq \frac{1 + \frac{7}{n^3}}{1} \leq 1 + 7 = 8,$$

hence

$$|a_n| \;=\; \frac{1}{n} \left| \frac{1 - \frac{7}{n^3}}{1 + \frac{3}{n^4}} \right| \;\leq\; \frac{8}{n}.$$

Given $\varepsilon > 0$, choose $N > \frac{8}{\varepsilon}$. Then for all $n \geq N$,

$$|a_n| \leq \frac{8}{n} \leq \frac{8}{N} < \varepsilon,$$

so indeed $a_n \to 0$.

Conclusion

$$\boxed{\lim_{n \to \infty} a_n = 0.}$$

□

(b)

Solution. Let

$$b_n := \frac{2^n}{n!} \qquad (n \in \mathbb{N}).$$

We show $b_n \to 0$.

Ratio step and geometric domination. Compute

$$\frac{b_{n+1}}{b_n} = \frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1}.$$

For $n \geq 3$ we have $\dfrac{2}{n+1} \leq \frac{1}{2}$, hence

$$b_{n+1} \leq \tfrac{1}{2}\, b_n \qquad (n \geq 3).$$

Iterating yields, for every $m > n \geq 3$,

$$b_m \leq b_n \left(\tfrac{1}{2}\right)^{m-n}.$$

Taking $n = 3$ (where $b_3 = \dfrac{2^3}{3!} = \dfrac{4}{3}$) gives

$$b_m \leq \frac{4}{3}\left(\tfrac{1}{2}\right)^{m-3} = \frac{32}{3}\left(\tfrac{1}{2}\right)^{m},$$

so the sequence is eventually dominated by a geometric sequence with ratio 1/2. Hence $b_n \to 0$.

Epsilon proof from the bound. From the previous bound,

$$0 \leq b_n \leq \frac{32}{3}\left(\tfrac{1}{2}\right)^{n} \qquad (n \geq 3).$$

Given $\varepsilon > 0$, choose N with $\dfrac{32}{3}\left(\frac{1}{2}\right)^{N} < \varepsilon$. Then for all $n \geq N$,

$$|b_n - 0| = b_n \leq \frac{32}{3}\left(\tfrac{1}{2}\right)^{n} \leq \frac{32}{3}\left(\tfrac{1}{2}\right)^{N} < \varepsilon,$$

proving $b_n \to 0$.

Remarks. The ratio $\dfrac{b_{n+1}}{b_n} = \dfrac{2}{n+1} \to 0$ encodes that $n!$ outgrows 2^n. Alternatively, since $e^2 = \sum_{k=0}^{\infty} \frac{2^k}{k!}$ converges, its terms satisfy $\dfrac{2^n}{n!} \to 0$. □

(c)

Solution. Let

$$c_n := \frac{n^2}{2^n + 3^n}.$$

We show $c_n \to 0$ by bounding c_n with a simpler sequence and applying geometric domination.

Step 1: Simplify using a lower bound on the denominator. Since $3^n \leq 2^n + 3^n$, we have

$$0 \leq c_n = \frac{n^2}{2^n + 3^n} \leq \frac{n^2}{3^n}.$$

Thus it suffices to prove

$$d_n := \frac{n^2}{3^n} \longrightarrow 0.$$

Step 2: Compute the ratio d_{n+1}/d_n. By definition,

$$d_n = \frac{n^2}{3^n}, \qquad d_{n+1} = \frac{(n+1)^2}{3^{n+1}}.$$

Dividing yields

$$\frac{d_{n+1}}{d_n} = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{(n+1)^2}{n^2} \cdot \frac{1}{3} = \left(1 + \frac{1}{n}\right)^2 \cdot \frac{1}{3}.$$

Step 3: Bound the ratio for $n \geq 2$. Expanding the square,

$$\left(1 + \frac{1}{n}\right)^2 = 1 + \frac{2}{n} + \frac{1}{n^2}.$$

For $n \geq 2$, we have $\frac{2}{n} \leq 1$ and $\frac{1}{n^2} \leq \frac{1}{4}$, so

$$\left(1 + \frac{1}{n}\right)^2 \leq 1 + 1 + \frac{1}{4} = \frac{9}{4}.$$

Hence

$$\frac{d_{n+1}}{d_n} \leq \frac{9}{4} \cdot \frac{1}{3} = \frac{3}{4} \qquad (n \geq 2),$$

or equivalently,

$$d_{n+1} \leq \frac{3}{4} d_n \qquad (n \geq 2).$$

Step 4: Geometric domination by iteration. Iterating the inequality for any $m > n \geq 2$ gives

$$d_m \leq d_n \left(\frac{3}{4}\right)^{m-n}.$$

Taking $n = 2$ (where $d_2 = \frac{4}{9}$), we obtain for all $m \geq 2$,

$$d_m \leq \frac{4}{9} \left(\frac{3}{4}\right)^{m-2} = \frac{64}{9} \left(\frac{3}{4}\right)^m.$$

Since $\left(\frac{3}{4}\right)^m \to 0$, we conclude $d_m \to 0$.

Step 5: Conclude $c_n \to 0$. Since $0 \leq c_n \leq d_n$ and $d_n \to 0$, by the squeeze theorem,

$$\boxed{\lim_{n\to\infty} c_n = 0.}$$

Optional ε-proof. From the geometric bound,

$$0 \leq c_n \leq \frac{64}{9} \left(\frac{3}{4}\right)^n \qquad (n \geq 2).$$

Given $\varepsilon > 0$, choose N such that $\frac{64}{9} \left(\frac{3}{4}\right)^N < \varepsilon$. Then for all $n \geq N$,

$$|c_n - 0| = c_n \leq \frac{64}{9} \left(\frac{3}{4}\right)^n \leq \frac{64}{9} \left(\frac{3}{4}\right)^N < \varepsilon,$$

proving $c_n \to 0$. $\square$

(d)

Solution. Let

$$d_n := \sqrt{n+1} - \sqrt{n}.$$

We show step by step that $d_n \to 0$ and give a clean bound and an ε-proof.

1) Rationalize the difference.

Multiply by the conjugate:

$$\begin{aligned} d_n &= \sqrt{n+1} - \sqrt{n} \\ &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{(\sqrt{n+1})^2 - (\sqrt{n})^2}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} \\ &= \boxed{\frac{1}{\sqrt{n+1} + \sqrt{n}}}. \end{aligned}$$

2) Bound it between two simple expressions.

Since $\sqrt{n+1} \geq \sqrt{n}$,

$$\sqrt{n+1} + \sqrt{n} \geq 2\sqrt{n} \implies d_n \leq \frac{1}{2\sqrt{n}}.$$

Also,

$$\sqrt{n+1} + \sqrt{n} \leq 2\sqrt{n+1} \implies d_n \geq \frac{1}{2\sqrt{n+1}}.$$

So,

$$\boxed{\frac{1}{2\sqrt{n+1}} \leq d_n \leq \frac{1}{2\sqrt{n}}}$$

Both bounds tend to 0, so by the squeeze theorem,

$$\lim_{n\to\infty} d_n = 0.$$

3) ε-proof (from the bound).

From above,

$$0 \leq d_n \leq \frac{1}{2\sqrt{n}}.$$

Given $\varepsilon > 0$, choose N such that

$$\frac{1}{2\sqrt{N}} < \varepsilon \iff N > \frac{1}{4\varepsilon^2}.$$

Then for all $n \geq N$,

$$|d_n - 0| = d_n \leq \frac{1}{2\sqrt{n}} \leq \frac{1}{2\sqrt{N}} < \varepsilon,$$

so $d_n \to 0$.

4) Optional: monotonicity.

Since

$$d_n = \frac{1}{\sqrt{n+1} + \sqrt{n}},$$

the denominator increases with n, so $d_{n+1} \leq d_n$ for all n. Thus (d_n) is positive, decreasing, and tends to 0. $\square$

2

(a)

Given:

- $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are sequences of real numbers
- $a_n \to \infty$
- $b_n \to b \in \mathbb{R}$

To prove: $a_n + b_n \to \infty$

Solution

Step 1: Write out what we need to prove

By definition, we need to show that $a_n + b_n \to \infty$, which means:

$$\forall M > 0, \exists N \in \mathbb{N} : \forall n \geq N, \quad a_n + b_n > M$$

Step 2: Use the given information

Since $a_n \to \infty$, we know:

$$\forall M' > 0, \exists N_1 \in \mathbb{N} : \forall n \geq N_1, \quad a_n > M'$$

Since $b_n \to b \in \mathbb{R}$, we know:

$$\forall \varepsilon > 0, \exists N_2 \in \mathbb{N} : \forall n \geq N_2, \quad |b_n - b| < \varepsilon$$

Step 3: Unpack the convergence condition for b_n

From $|b_n - b| < \varepsilon$, we get:

$$-\varepsilon < b_n - b < \varepsilon$$

Adding b to all parts:

$$b - \varepsilon < b_n < b + \varepsilon$$

In particular, we have:

$$b_n > b - \varepsilon$$

Step 4: Combine the estimates

For $n \geq \max\{N_1, N_2\}$, we have both $a_n > M'$ and $b_n > b - \varepsilon$.

Adding these inequalities:

$$a_n + b_n > M' + (b - \varepsilon) = M' + b - \varepsilon$$

Step 5: Choose appropriate values

Let $M > 0$ be given (arbitrary).

Choose $\varepsilon = 1$ (any positive constant works).

Choose $M' = M - b + 1$.

Then:

$$a_n + b_n > M' + b - \varepsilon = (M - b + 1) + b - 1 = M$$

Step 6: Set N and conclude

Let $N = \max\{N_1, N_2\}$, where:

- N_1 is such that $a_n > M - b + 1$ for all $n \geq N_1$
- N_2 is such that $|b_n - b| < 1$ for all $n \geq N_2$

Then for all $n \geq N$:

$$a_n + b_n > M$$

Since M was arbitrary, this proves that $a_n + b_n \to \infty$. $\square$

(b)

Given:

- $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are sequences of real numbers
- $a_n \to \infty$
- $b_n \to b \in \mathbb{R}$

To prove:

- If $b > 0$, then $a_n b_n \to \infty$
- If $b < 0$, then $a_n b_n \to -\infty$

Solution. I'll prove both cases separately.

Case 1: $b > 0$

Goal: Show that $a_n b_n \to \infty$

Step 1: Write what we need to prove

We need to show:

$$\forall M > 0, \exists N \in \mathbb{N} : \forall n \geq N, \quad a_n b_n > M$$

Step 2: Use the convergence of b_n

Since $b_n \to b$ and $b > 0$, we can choose $\varepsilon = \frac{b}{2} > 0$.

Then $\exists N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$:

$$|b_n - b| < \frac{b}{2}$$

Step 3: Get a lower bound for b_n

From $|b_n - b| < \frac{b}{2}$:

$$-\frac{b}{2} < b_n - b < \frac{b}{2}$$

Adding b to all parts:

$$b - \frac{b}{2} < b_n < b + \frac{b}{2}$$

Simplifying:

$$\frac{b}{2} < b_n < \frac{3b}{2}$$

In particular:

$$b_n > \frac{b}{2} > 0$$

Step 4: Use the divergence of a_n

Since $a_n \to \infty$, for any $M' > 0$, $\exists N_2 \in \mathbb{N}$ such that $\forall n \geq N_2$:

$$a_n > M'$$

Step 5: Combine the estimates

Let $M > 0$ be given (arbitrary).

Choose $M' = \frac{2M}{b}$.

Then $\exists N_2$ such that for $n \geq N_2$:

$$a_n > \frac{2M}{b}$$

For $n \geq \max\{N_1, N_2\}$, we have both $a_n > \frac{2M}{b}$ and $b_n > \frac{b}{2}$.

Since both are positive, we can multiply:

$$a_n b_n > \frac{2M}{b} \cdot \frac{b}{2} = M$$

Step 6: Conclude

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$:

$$a_n b_n > M$$

Since M was arbitrary, $a_n b_n \to \infty$. $\square$

Case 2: $b < 0$

Goal: Show that $a_n b_n \to -\infty$

Step 1: Write what we need to prove

We need to show:

$$\forall M > 0, \exists N \in \mathbb{N} : \forall n \geq N, \quad a_n b_n < -M$$

(Note: For divergence to $-\infty$, we need the sequence to eventually be less than $-M$ for any positive M)

Step 2: Use the convergence of b_n

Since $b_n \to b$ and $b < 0$, we can choose $\varepsilon = \frac{|b|}{2} = -\frac{b}{2} > 0$.

Then $\exists N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$:

$$|b_n - b| < -\frac{b}{2}$$

Step 3: Get an upper bound for b_n

From $|b_n - b| < -\frac{b}{2}$:

$$\frac{b}{2} < b_n - b < -\frac{b}{2}$$

Adding b to all parts:

$$b + \frac{b}{2} < b_n < b - \frac{b}{2}$$

Simplifying:

$$\frac{3b}{2} < b_n < \frac{b}{2}$$

In particular:

$$b_n < \frac{b}{2} < 0$$

Step 4: Use the divergence of a_n

Since $a_n \to \infty$, for any $M' > 0$, $\exists N_2 \in \mathbb{N}$ such that $\forall n \geq N_2$:

$$a_n > M'$$

Step 5: Combine the estimates

Let $M > 0$ be given (arbitrary).

Choose $M' = \frac{2M}{|b|} = -\frac{2M}{b}$ (note: this is positive since $b < 0$).

Then $\exists N_2$ such that for $n \geq N_2$:

$$a_n > -\frac{2M}{b}$$

For $n \geq \max\{N_1, N_2\}$, we have $a_n > -\frac{2M}{b} > 0$ and $b_n < \frac{b}{2} < 0$.
Multiplying (inequality reverses since $b_n < 0$):

$$a_n b_n < -\frac{2M}{b} \cdot \frac{b}{2} = -M$$

Step 6: Conclude

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$:

$$a_n b_n < -M$$

Since M was arbitrary, $a_n b_n \to -\infty$. □ □

3

(a)

Solution. The statement is false. We will disprove it by providing a counterexample.
Let

$$a_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

Then

$$b_n = a_{n+1} - a_n = \frac{1}{n+1}.$$

We will show that $b_n \to 0$ but a_n does not converge.

Step 1: Show that $b_n \to 0$

We have

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n+1} = 0.$$

To prove this, let $\varepsilon > 0$ be given. We want to find $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\left| \frac{1}{n+1} - 0 \right| < \varepsilon.$$

This simplifies to

$$\frac{1}{n+1} < \varepsilon,$$

which is equivalent to

$$n + 1 > \frac{1}{\varepsilon},$$

or

$$n > \frac{1}{\varepsilon} - 1.$$

Choose $N = \lceil \frac{1}{\varepsilon} \rceil$. Then for all $n \geq N$,

$$n \geq N \geq \frac{1}{\varepsilon} - 1,$$

so

$$n + 1 > \frac{1}{\varepsilon},$$

and thus

$$\frac{1}{n+1} < \varepsilon.$$

Therefore, $b_n \to 0$.

Step 2: Show that a_n does not converge

The sequence $a_n = \sum_{k=1}^{n} \frac{1}{k}$ is the harmonic series, which is known to diverge to infinity. We will show that $a_n \to \infty$.

Write out the terms:

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \cdots + \frac{1}{n}.$$

Group them as follows:

$$a_n = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \ldots$$

Estimate each group from below:

- First group: $1 = 1$
- Second group: $\frac{1}{2} = \frac{1}{2}$
- Third group: $\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$
- Fourth group: $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$

Each group (after the first) contributes more than $\frac{1}{2}$ to the sum.

For $n = 2^k$, we have approximately k groups, so

$$a_{2^k} > 1 + \frac{k}{2}.$$

As $k \to \infty$, we have $\frac{k}{2} \to \infty$, so $a_{2^k} \to \infty$.

Since (a_n) has a subsequence that diverges to infinity, the sequence a_n itself diverges to infinity:

$$a_n \to \infty.$$

Therefore, a_n does not converge to any finite limit.

Step 3: Conclusion

We have shown that $b_n = a_{n+1} - a_n = \frac{1}{n+1} \to 0$, but $a_n = \sum_{k=1}^{n} \frac{1}{k} \to \infty$ (does not converge). This provides a counterexample to the claim, proving that the statement is false. $\square$

Alternative 3a):

Solution. **(a)** The statement is false. Consider the sequence $a_n = \sqrt{n}$. Then

$$b_n = a_{n+1} - a_n = \sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

Since $\sqrt{n+1} + \sqrt{n} \to \infty$ as $n \to \infty$, we have $b_n \to 0$. However, $a_n = \sqrt{n} \to \infty$ as $n \to \infty$, so a_n diverges. □

(b)

Solution. Assume $b_n := a_{n+1} - a_n \to 0$. We prove that $\dfrac{a_n}{n} \to 0$.

1) Telescoping representation and division by n. For every $n \geq 2$,

$$a_n - a_1 = \sum_{k=1}^{n-1} (a_{k+1} - a_k) = \sum_{k=1}^{n-1} b_k,$$

hence

$$\frac{a_n}{n} = \frac{a_1}{n} + \frac{1}{n} \sum_{k=1}^{n-1} b_k.$$

Clearly $\dfrac{a_1}{n} \to 0$, so it remains to show $\dfrac{1}{n} \sum_{k=1}^{n-1} b_k \to 0$.

2) ε-step and definition of C. Let $\varepsilon > 0$. Since $b_k \to 0$, there exists $N \in \mathbb{N}$ such that for all $k \geq N$, $|b_k| \leq \varepsilon$. Define the finite constant

$$C := \sum_{k=1}^{N-1} |b_k|.$$

3) Split the sum and bound via triangle inequality. For $n \geq N + 1$,

$$\begin{aligned}
\left| \frac{1}{n} \sum_{k=1}^{n-1} b_k \right| &= \frac{1}{n} \left| \sum_{k=1}^{N-1} b_k + \sum_{k=N}^{n-1} b_k \right| \\
&\leq \frac{1}{n} \left(\left| \sum_{k=1}^{N-1} b_k \right| + \left| \sum_{k=N}^{n-1} b_k \right| \right) && \text{triangle inequality} \\
&\leq \frac{1}{n} \left(\sum_{k=1}^{N-1} |b_k| + \sum_{k=N}^{n-1} |b_k| \right) && \text{triangle inequality on sums} \\
&\leq \frac{1}{n} \left(C + \sum_{k=N}^{n-1} \varepsilon \right) && (|b_k| \leq \varepsilon \text{ for } k \geq N) \\
&= \frac{1}{n} \left(C + \varepsilon \left(n - 1 - N + 1 \right) \right) \\
&= \frac{C}{n} + \varepsilon \frac{n - N}{n} \\
&= \frac{C}{n} + \varepsilon \Big(1 - \frac{N}{n} \Big).
\end{aligned}$$

4) Final simplification of the factor $1 - \frac{N}{n}$. For $n \geq N$ we have $0 \leq \frac{N}{n} \leq 1$, hence $0 \leq 1 - \frac{N}{n} \leq 1$. Since $\varepsilon > 0$, multiplying preserves the inequalities:

$$0 \leq \varepsilon\Big(1 - \tfrac{N}{n}\Big) \leq \varepsilon.$$

Therefore,

$$\left|\frac{1}{n}\sum_{k=1}^{n-1} b_k\right| \leq \frac{C}{n} + \varepsilon.$$

5) Passage to the limit. Taking lim sup and using $\frac{C}{n} \to 0$ gives

$$\limsup_{n\to\infty}\left|\frac{1}{n}\sum_{k=1}^{n-1} b_k\right| \leq \varepsilon.$$

Because $\varepsilon > 0$ is arbitrary, we conclude $\dfrac{1}{n}\sum_{k=1}^{n-1} b_k \to 0$. Combining with $\dfrac{a_1}{n} \to 0$ yields

$$\boxed{\lim_{n\to\infty}\frac{a_n}{n} = 0}.$$

Optional one-liner (Stolz–Cesàro). With $c_n = n$ (strictly increasing, $c_n \to \infty$),

$$\lim_{n\to\infty}\frac{a_n}{n} = \lim_{n\to\infty}\frac{a_{n+1} - a_n}{(n+1) - n} = \lim_{n\to\infty} b_n = 0.$$

□

4

(a)

Solution. **Goal.** Show that $(0,1)$ and $\mathbb{R}$ have the same cardinality by constructing an explicit bijection.

Definition of the map. Define

$$f : (0,1) \to \mathbb{R}, \qquad f(x) = \tan\big(\pi(x - \tfrac{1}{2})\big).$$

Step 1: Well-definedness. For $x \in (0,1)$ we have $\theta(x) := \pi(x - \frac{1}{2}) \in (-\frac{\pi}{2}, \frac{\pi}{2})$, where tan is finite and continuous. Hence $f(x) \in \mathbb{R}$ for all $x \in (0,1)$.

Step 2: Injectivity (strictly increasing). Differentiate:

$$f'(x) = \pi\,\sec^2\big(\pi(x - \tfrac{1}{2})\big) = \pi\big(1 + \tan^2(\pi(x - \tfrac{1}{2}))\big) > 0,$$

so f is strictly increasing on $(0,1)$ and therefore injective.

Step 3: Surjectivity (endpoint limits). As $x \to 0^+$, $\theta(x) \to (-\frac{\pi}{2})^+$ and $\tan\theta \to -\infty$. As $x \to 1^-$, $\theta(x) \to (\frac{\pi}{2})^-$ and $\tan\theta \to +\infty$. Because f is continuous and strictly

increasing on $(0,1)$ and its image runs from $-\infty$ to $+\infty$, f hits every real value exactly once; thus f is surjective.

Step 4: Explicit inverse and two-sided verification. Define

$$f^{-1} : \mathbb{R} \to (0,1), \qquad f^{-1}(y) = \frac{1}{\pi}\Big(\arctan(y) + \frac{\pi}{2}\Big).$$

Since $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $\arctan(y) + \frac{\pi}{2} \in (0,\pi)$ and hence $f^{-1}(y) \in (0,1)$. Moreover, for $x \in (0,1)$, $\theta(x) = \pi(x - \frac{1}{2}) \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so

$$f^{-1}(f(x)) = \frac{1}{\pi}\big(\arctan(\tan\theta(x)) + \tfrac{\pi}{2}\big) = \frac{1}{\pi}(\theta(x) + \tfrac{\pi}{2}) = x,$$

and for all $y \in \mathbb{R}$,

$$f(f^{-1}(y)) = \tan\Big(\arctan(y)\Big) = y.$$

Thus f is a bijection with inverse f^{-1}.

Trig identities/facts used.

- $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is bijective with inverse $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$; consequently $\tan(\arctan y) = y$ for all $y \in \mathbb{R}$ and $\arctan(\tan\theta) = \theta$ for $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.
- Monotonicity: $\dfrac{d}{d\theta}\tan\theta = \sec^2\theta = 1 + \tan^2\theta > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- Endpoint limits: $\lim_{\theta\to(\pi/2)^-} \tan\theta = +\infty$ and $\lim_{\theta\to(-\pi/2)^+} \tan\theta = -\infty$ (since $\tan = \sin / \cos$ with $\cos \to 0^{\pm}$).

$$\boxed{f(x) = \tan\big(\pi(x - \tfrac{1}{2})\big) \text{ is a bijection } (0,1) \leftrightarrow \mathbb{R},\ f^{-1}(y) = \tfrac{1}{\pi}(\arctan y + \tfrac{\pi}{2})}.$$

Optional alternative. The map $h(x) = \ln\big(\frac{x}{1-x}\big)$ also defines a bijection $(0,1) \to \mathbb{R}$ with inverse $h^{-1}(t) = \dfrac{e^t}{1+e^t}$. □

5

(a)

Solution. **What "algebraic" means.** A real number x is called *algebraic* if there exist integers $a_0, \ldots, a_n$, not all zero, such that

$$a_0 + a_1 x + \cdots + a_n x^n = 0.$$

Equivalently, x is a root of some nonzero polynomial with integer (or rational) coefficients. If a polynomial with rational coefficients vanishes at x, clearing denominators yields an integer-coefficient polynomial that also vanishes at x.

Claim. The set $\mathcal{P}$ of polynomials with integer coefficients is countable.

Facts used about countability.

1) $\mathbb{Z}$ is countable.

2) A finite Cartesian product of countable sets is countable.

3) A countable union of countable sets is countable.

Step 1: Fix the degree. For a fixed $n \in \mathbb{N}$, the set of coefficient tuples

$$\mathbb{Z}^{n+1} = \{(a_0, \ldots, a_n) : a_i \in \mathbb{Z}\}$$

is countable by 1) and 2). Define the surjection

$$\Phi_n : \mathbb{Z}^{n+1} \to \{\text{polynomials with integer coefficients of degree } \leq n\},$$

$$(a_0, \ldots, a_n) \mapsto a_0 + a_1 X + \cdots + a_n X^n.$$

The image of a countable set is countable, so the set of polynomials of degree $\leq n$ is countable.

Step 2: Union over all degrees. Let $\mathcal{P}_{\leq n}$ denote the set of integer-coefficient polynomials of degree $\leq n$. Then

$$\mathcal{P} = \bigcup_{n=0}^{\infty} \mathcal{P}_{\leq n}$$

is a countable union of countable sets, hence countable by 3).

Optional. Excluding the zero polynomial merely removes one element and preserves countability.

Conclusion.

$$\boxed{\text{The set of polynomials with integer coefficients is countable.}}$$

□

(b)

Solution. **Claim.** The set $\mathcal{A}$ of algebraic real numbers is countable.

Reasoning. For each nonzero polynomial $p \in \mathcal{P}$ with integer coefficients, the set of its real roots is finite (in fact of size at most $\deg p$ by the fundamental theorem of algebra and the nonzero-derivative/root isolation fact that a nonzero real polynomial of degree n has at most n real roots). Let

$$R(p) := \{x \in \mathbb{R} : p(x) = 0\}.$$

Then $R(p)$ is finite for each $p \neq 0$; hence $R(p)$ is countable. From part (a), the index set $\mathcal{P} \setminus \{0\}$ is countable. Therefore

$$\mathcal{A} = \bigcup_{p \in \mathcal{P} \setminus \{0\}} R(p)$$

is a countable union of countable (indeed finite) sets, so $\mathcal{A}$ is countable.

$$\boxed{\text{The set of algebraic real numbers is countable.}}$$

□

(c)

Solution. Since $\mathbb{R}$ is uncountable (Cantor's diagonal argument) while the set of algebraic reals $\mathcal{A}$ is countable by part (b), the complement

$$\mathbb{R} \setminus \mathcal{A}$$

is nonempty (indeed uncountable). Any element of this complement is a real number that is not algebraic, i.e., a transcendental real.

$$\boxed{\text{There exists a real number that is not algebraic (transcendental).}}$$

$\square$